1. Project Title

PPG-Based Cardiovascular and Respiratory Signal Analysis Using ESP32, MAX30102, and MATLAB

2. Objective

The objective of this project is to develop a low-cost biomedical signal acquisition and analysis system using **ESP32**, **MAX30102**, and **MATLAB**. The system is designed to collect **PPG (Photoplethysmography)** signals from the fingertip and process them to extract useful physiological information.

The specific objectives of this project are:

1. To estimate **arterial stiffness / vascular aging trend** using the **second derivative of the PPG signal**.
2. To estimate **stress trend** from **pulse rate variability (PRV)** obtained from pulse intervals.
3. To estimate **respiration rate** from the **slow respiratory modulation present in the PPG signal**.
4. To perform **irregular pulse screening** using **pulse interval variability analysis and Poincaré plot representation**.

3. Equipment List

The following equipment and software are used in this project:

Hardware

- **ESP32 ESP-32S 30-pin NodeMCU Development Board**
- **MAX30102 PPG Sensor Module**
- **Jumper wires**
- **USB cable**
- **Laptop / PC**

Software

- **Arduino IDE 2.3.8**
- **MATLAB R2020a**

4. Working Principle

This project is based on the principle of **Photoplethysmography (PPG)**. The **MAX30102** sensor emits light into the fingertip and detects the reflected light intensity. Since blood volume inside the finger changes with every heartbeat, the amount of reflected light also changes. These changes form a **PPG waveform**, which contains useful cardiovascular and respiratory information.

The **ESP32** reads the PPG data from the MAX30102 through **I2C communication** and sends the signal data to the computer through serial communication. The recorded data is saved in text format and then processed in **MATLAB**.

The overall working procedure of the project is as follows:

Step 1: Signal Acquisition

The MAX30102 sensor is placed under the finger to acquire the PPG signal. The ESP32 collects the signal continuously.

Step 2: Data Transmission

The ESP32 sends the recorded PPG data to the computer through serial communication.

Step 3: MATLAB Signal Processing

The recorded signal is imported into MATLAB, where different signal processing techniques are applied, such as:

- smoothing
- filtering
- peak detection
- pulse interval extraction
- derivative analysis
- spectral analysis
- variability analysis

Step 4: Physiological Parameter Estimation

From the processed PPG signal, the following analyses are performed:

1) Arterial stiffness / vascular age trend

Show:

- averaged PPG pulse plot
- first derivative plot
- second derivative plot
- marked **a, b, c, d, e** points
- final values:
    - **b/a**
    - **c/a**
    - **d/a**
    - **e/a**
    - **AGI**

2) Stress trend from PRV

Show:

- detected pulse peaks plot
- pulse interval plot
- PRV power spectral density plot
- final values:
    - **SDNN**
    - **RMSSD**
    - **LF**
    - **HF**
    - **LF/HF**
- final interpretation:
    - relaxed / mild stress / higher stress trend

3) Respiration rate

Show:

- raw PPG plot
- extracted respiratory modulation plot
- respiration frequency spectrum
- final value:
    - **respiration rate in breaths/min**

4) Irregular pulse screening

Show:

- pulse interval plot
- absolute successive interval change plot
- **Poincaré plot**
- final values:
    - **SD1**
    - **SD2**
    - **SD1/SD2**
    - **irregular interval %**
- final interpretation:
    - no strong irregular pulse trend / irregular pulse trend detected

Thus, the complete system converts raw optical pulse data into meaningful physiological indicators through digital signal processing.

5. Discussion

This project is highly relevant because cardiovascular and respiratory monitoring plays an important role in modern healthcare, personal wellness monitoring, and biomedical engineering research. Parameters such as pulse rate, stress-related variability, breathing rate, and pulse irregularity can provide valuable information about the physiological condition of a person.

Many medical monitoring devices available in the market are costly and may not always be suitable for low-cost educational, research, or prototype development purposes. In contrast, this project demonstrates that a comparatively simple and low-cost setup using **ESP32**, **MAX30102**, and **MATLAB** can be used to perform meaningful biomedical signal analysis.

This project can help solve real-life problems in several ways:

- It can support **basic physiological monitoring** using affordable hardware.
- It can provide **stress trend estimation** through pulse interval variability analysis.
- It can estimate **respiration rate** without requiring a separate breathing sensor.
- It can perform **irregular pulse screening**, which may be useful for early observation of abnormal pulse behavior.
- It can serve as a **practical prototype for wearable or portable health monitoring systems**.
- It has strong educational value for understanding **biomedical sensors, microcontroller interfacing, and signal processing techniques**.

Although this system is not intended to replace clinical diagnostic equipment, it is very useful for **screening, trend observation, academic demonstration, and research-based biomedical**

engineering projects. Therefore, this project is both technically meaningful and practically relevant.

Hardware Setup

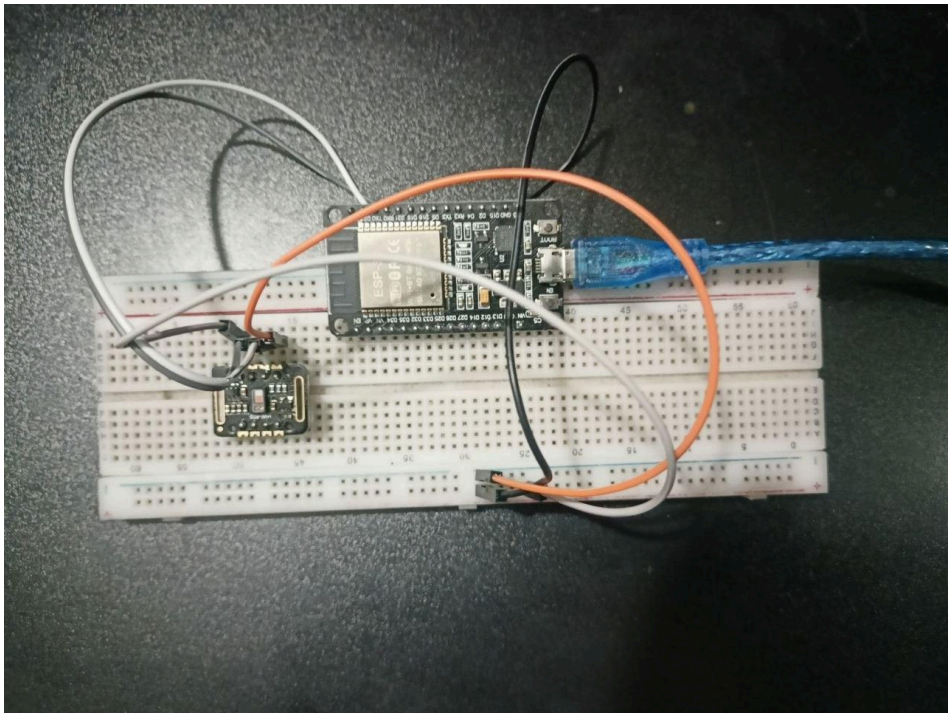

Figure 1: Real-Time Acquisition of PPG Signal from MAX30102 Using ESP32

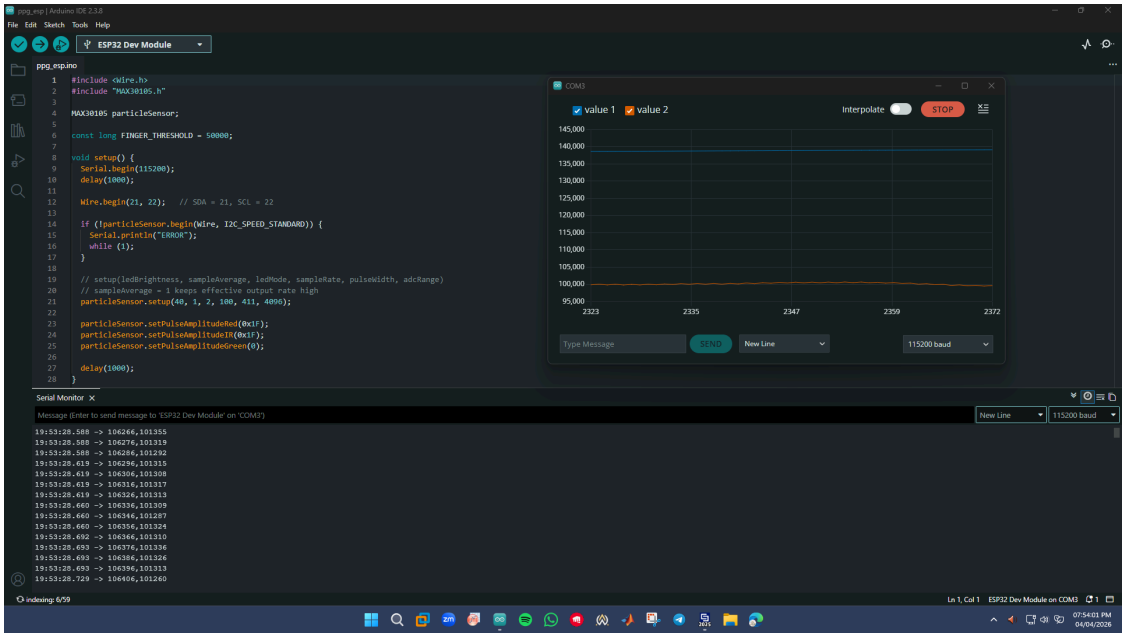

Figure 2: Recorded Raw Photoplethysmography (PPG) Signal

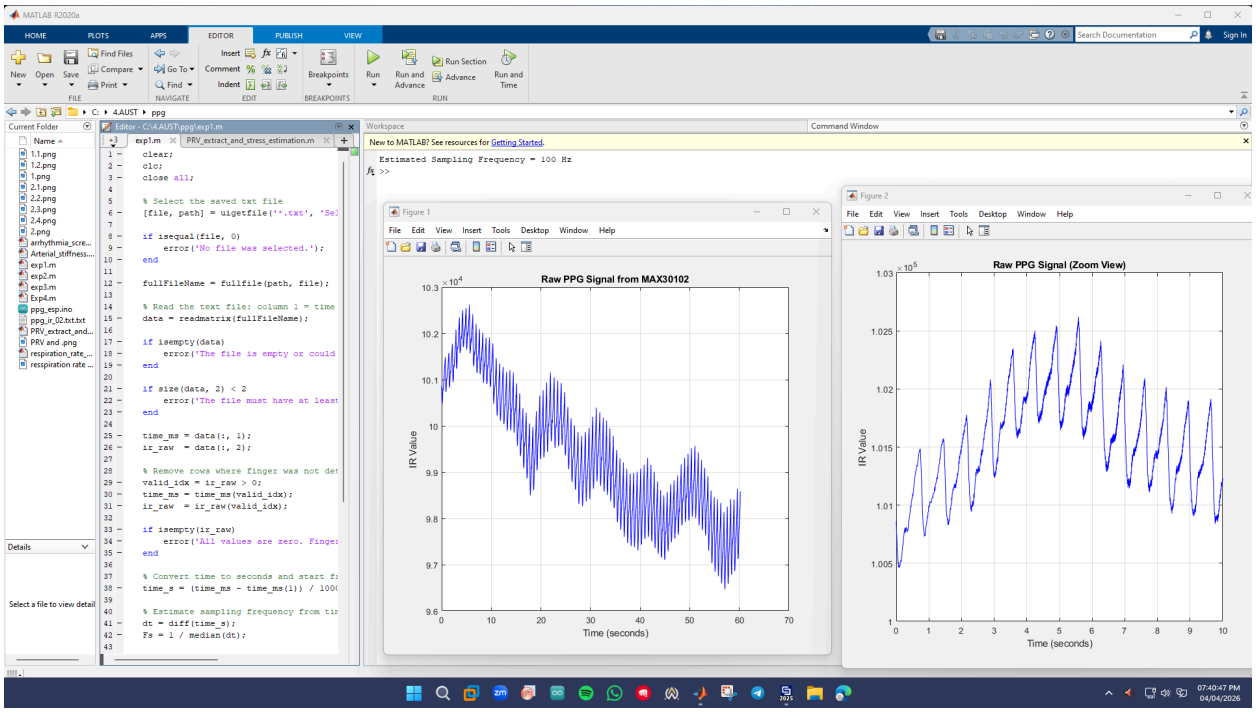

Figure 3: Extraction of Respiratory Information from the PPG Signal

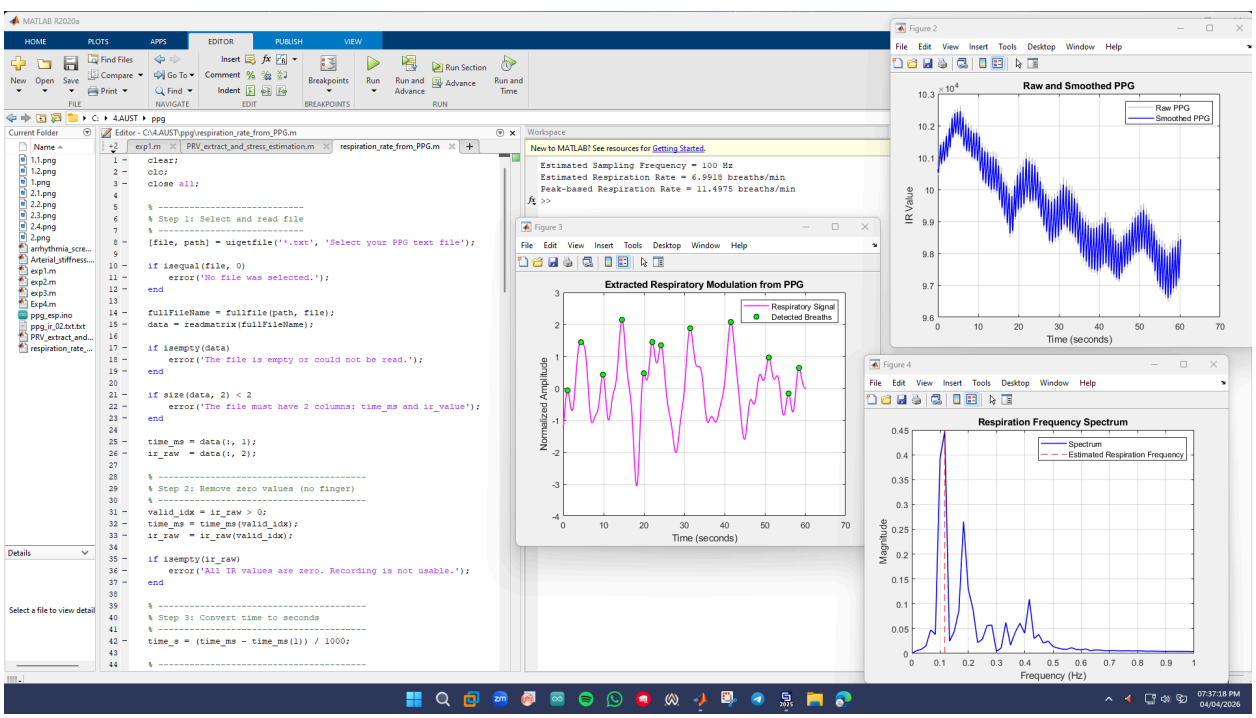

Figure 4: Pulse Rate Variability (PRV) Analysis for Stress Estimation

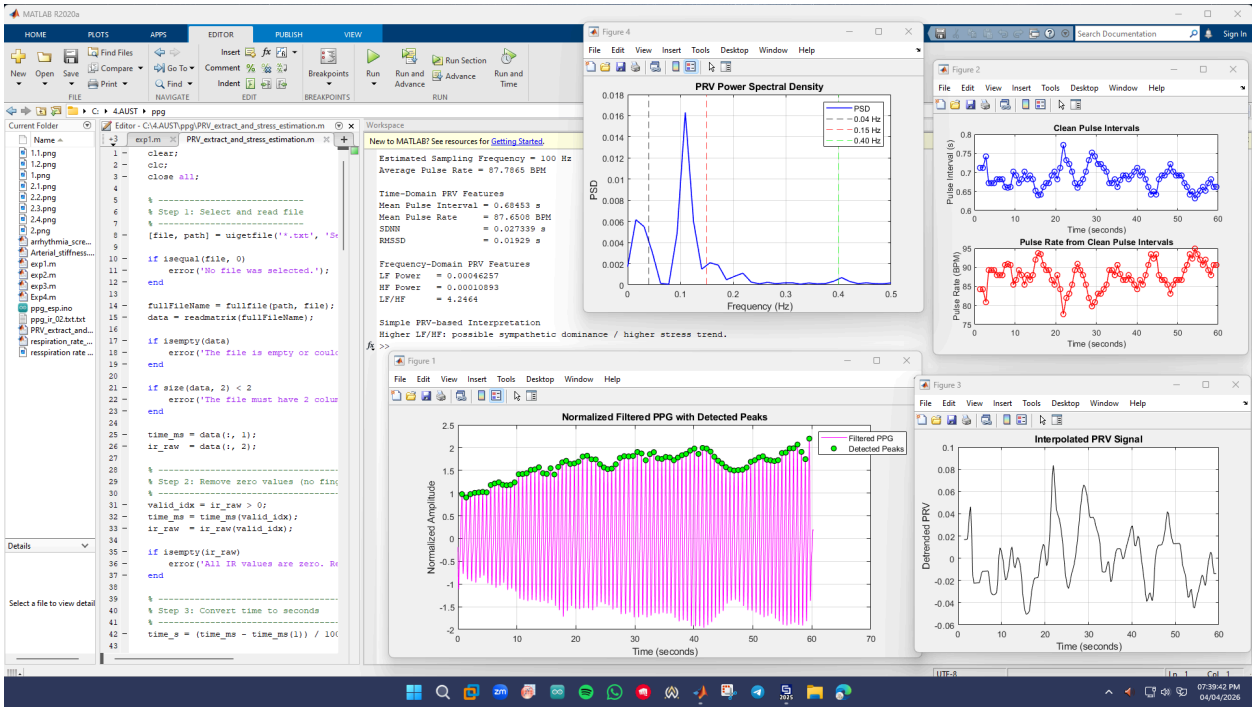

Figure 5: Arterial Stiffness Assessment from the Second Derivative of PPG

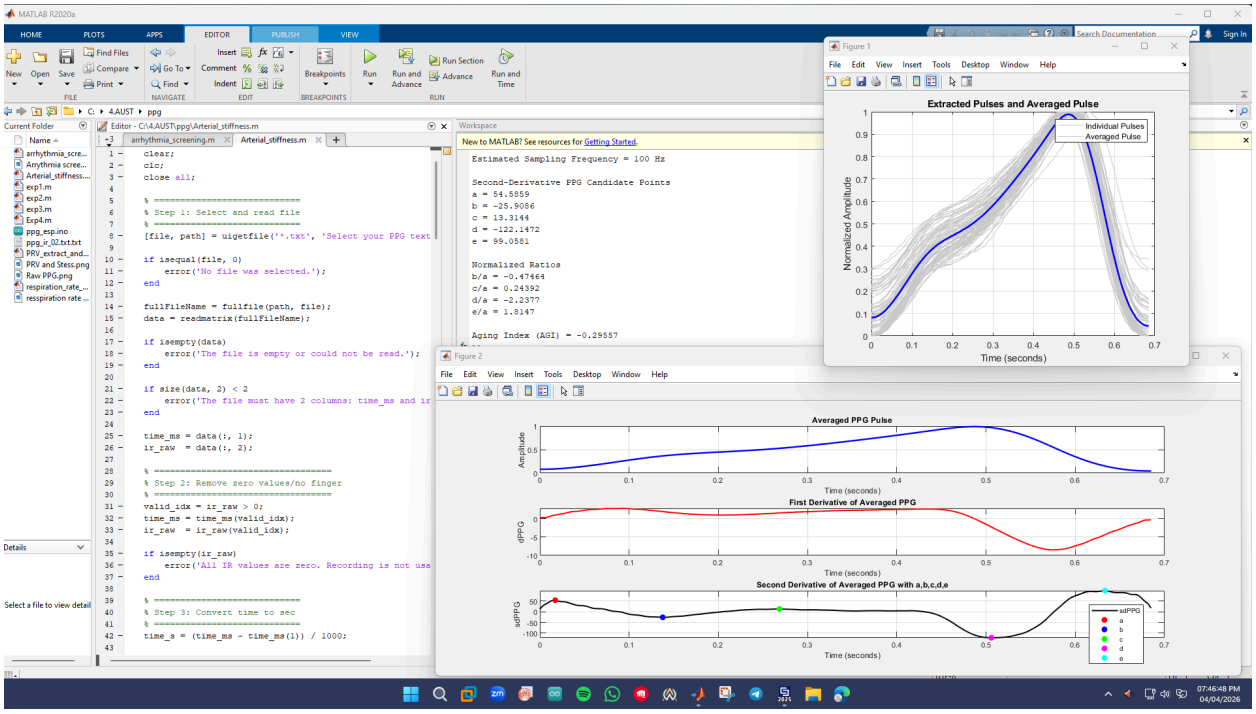
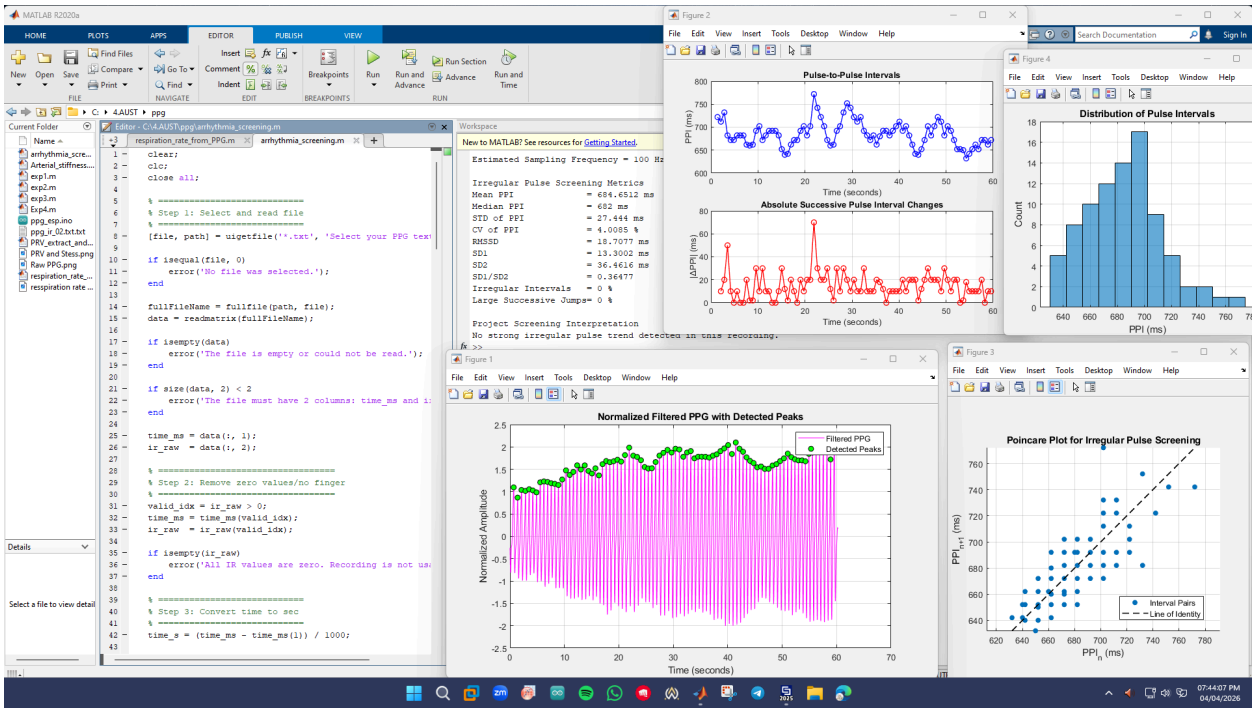
Figure 6: Irregular Pulse Screening and Arrhythmia Detection from PPG